

[Back to Index](#)

[FIND EVIL! Hackathon](#)

Table of Contents

- 1. [Category 1: Autonomous DFIR / IR Agents](#) (6 papers)
- 2. [Category 2: AI Agent Architecture for Security](#) (5 papers)
- 3. [Category 3: Defensive AI](#) (4 papers)
- 4. [Category 4: AI Offensive/Defensive Balance](#) (4 papers)
- 5. [Category 5: Forensic Automation](#) (6 papers)
- 6. [Supplementary Resources](#)
- 7. [Architecture Implications: Synthesis](#)

Category 1: Autonomous DFIR / IR Agents

#	Title	Source	Date	Key Finding	URL
1	Chances and Challenges of MCP in DFIR	arXiv 2506.00274	May 2025	MCP as integration layer for LLMs across forensic workflows. Required reading for MCP server architecture.	https://arxiv.org/abs/2506.00274
2	CyberSleuth: Autonomous Blue-Team LLM Agent	arXiv 2508.20643	Aug 2025	Closest existing system to FIND EVIL. Multi-agent specialization outperforms single-agent. 80% accuracy on 2025 incidents.	https://arxiv.org/abs/2508.20643
3	Advancing Autonomous Incident Response	arXiv 2508.10677	Aug 2025	RAG-based framework pairing LLMs with CTI using STIX/TAXII.	https://arxiv.org/abs/2508.10677
4	DFIR-Metric Benchmark	arXiv 2505.19973	May 2025	First rigorous benchmark: 700 questions, 150 CTF challenges, 500 disk/memory cases from NIST CFTT.	https://arxiv.org/abs/2505.19973
5	Digital Forensics in the Age of LLMs	arXiv 2504.02963	Apr 2025	Systematic survey of LLM across full forensic pipeline.	https://arxiv.org/abs/2504.02963
6	SoK: LLMs for Digital Forensic Investigation	arXiv 2402.19366	Feb 2024	Foundational SoK paper establishing LLM applicability across forensic domains.	https://arxiv.org/abs/2402.19366

Category 2: AI Agent Architecture for Security

#	Title	Source	Date	Key Finding	URL
7	AI-Augmented SOC Survey	MDPI Journal of Cybersecurity and Privacy	Nov 2025	500+ papers reviewed, five-level Capability Maturity Model for AI-augmented SOC's.	https://www.mdpi.com/2624-800X/5/4/95

#	Title	Source	Date	Key Finding	URL
8	Agentic AI and Cybersecurity Survey	arXiv 2601.05293	Jan 2026	Transition from single-step to full agentic systems with memory/tool use.	https://arxiv.org/abs/2601.05293
9	Multi-Agent LLM Orchestration for IR	ResearchGate	2025	348 trials: multi-agent 100% actionable vs single-agent 1.7%.	https://www.researchgate.net/publication/397824192
10	Hyper-Automation SOAR with Agentic AI	MDPI Information	Apr 2025	Dynamic playbook generation vs static playbooks -- agentic wins on novel incident types.	https://www.mdpi.com/2078-2489/16/5/365
11	Policy-Guided Threat Hunting with Splunk SOC	arXiv 2603.23966	Mar 2026	CrewAI multi-agent with specialist roles outperforms monolithic agent on structured hunt tasks.	https://arxiv.org/abs/2603.23966

Category 3: Defensive AI

#	Title	Source	Date	Key Finding	URL
12	Project Glasswing / Claude Mythos	Anthropic	Apr 7, 2026	100M USD initiative, thousands of zero-days found through autonomous vulnerability research.	https://www.anthropic.com/glasswing
13	volGPT: Memory Forensics with LLMs	DFRWS 2024	2024	GPT + Volatility integration, 87-99% ransomware detection accuracy on memory artifacts.	https://www.sciencedirect.com/science/article/pii/S2666281724000751
14	ForensicLLM	DFRWS EU 2025	2025	Fine-tuned LLaMA-3.1-8B on 6,739 forensic Q&A pairs. 86.6% source attribution accuracy.	https://dfrws.org/presentation/forensicllm-a-local-large-language-model-for-digital-forensics/
15	LLMs for Memory Forensics Comparative Analysis	ACM	2025	Reasoning-enabled LLMs outperform standard	https://dl.acm.org/doi/full/10.1145/3748263

#	Title	Source	Date	Key Finding	URL
				models on memory forensic detection tasks.	

Category 4: AI Offensive/Defensive Balance

#	Title	Source	Date	Key Finding	URL
16	Agents of Chaos	arXiv 2602.20021	Feb 2026	11 failure case studies from live autonomous agent red-team operations. Already in vault: 03 Resources/Articles/2026-03-08_Agents-of-Chaos.	https://arxiv.org/abs/2602.20021
17	GTG-1002 Disruption Report	Anthropic	Nov 2025	Chinese state-sponsored AI exploitation documented at 80-90% autonomy. Attack request rates physically impossible for human operators.	https://assets.anthropic.com/m/ec212e6566a0d47/origin-the-first-reported-AI-orchestrated-cyber-espionage-campaign
18	PentestGPT	USENIX Security 2024	2024	Pentesting Task Tree for structured multi-step operations -- transferable to DFIR task sequencing.	https://www.usenix.org/conference/usenixsecurity24/presentation-2409.02572
19	What Makes a Good LLM Pentesting Agent	arXiv 2602.17622	Feb 2026	Empirical analysis of what separates high/low performing agents across structured security tasks.	https://arxiv.org/html/2602.17622

Category 5: Forensic Automation

#	Title	Source	Date	Key Finding	URL
20	GenDFIR: Timeline Analysis with Rule-Based AI + LLMs	arXiv 2409.02572	Sep 2024	Two-stage pipeline: rules-based triage then LLM reasoning. Reduces hallucination surface by grounding LLM on pre-filtered artifacts.	https://arxiv.org/abs/2409.02572
21	PROVSEEK: LLM Provenance Forensics	arXiv 2508.21323	Aug 2025	Provenance-driven forensics with PostgreSQL + ChromaDB + LangChain. Graph-based artifact linking.	https://arxiv.org/abs/2508.21323
22	Standardized Methodology for LLM Timeline Analysis	arXiv 2505.03100	May 2025	Evaluation framework for timeline reconstruction accuracy -- directly applicable to DFIR-Metric benchmarking.	https://arxiv.org/html/2505.03100v1
23	LLMs for CTI Workflow Automation	arXiv 2407.13093	Jul 2024	End-to-end CTI pipeline without human intervention. Covers ingestion, enrichment, correlation, reporting.	https://arxiv.org/html/2407.13093v1

#	Title	Source	Date	Key Finding	URL
24	LLM Agent Hallucination Survey	arXiv 2509.18970	Sep 2025	18 hallucination trigger causes specific to agents. Tool output grounding identified as most effective mitigation.	https://arxiv.org/abs/2509.18970
25	AIR: Agent Safety via Incident Response	arXiv 2602.11749	Feb 2026	Formal IR framework for agent systems -- defines rollback, containment, and safe degradation requirements.	https://arxiv.org/abs/2602.11749

Supplementary Resources

Resource	URL
SANS FOR563: Applied AI for DFIR	https://www.sans.org/cyber-security-courses/applied-ai-local-large-language-models
CoSAI MCP Security Taxonomy (OASIS, Jan 2026)	https://www.oasis-open.org/2026/01/27/coalition-for-secure-ai-releases-extensive-taxonomy-for-model-context-protocol-security/

Architecture Implications: Synthesis

Five evidence-backed design principles derived from the literature.

1. Multi-Agent Specialization Wins

Evidence

- CyberSleuth (paper 2): multi-agent specialization outperforms monolithic agent on 2025 incident types.
- Multi-Agent Orchestration (paper 9): 348 trials, 100% actionable output vs 1.7% for single-agent.
- Policy-Guided Threat Hunting (paper 11): CrewAI specialist roles outperform generalist agents.

Build the agent as a coordinator + specialists (triage, memory forensics, timeline reconstruction, CTI enrichment). Do not route all tasks through one model instance.

2. Dynamic Hypothesis-Driven Tool Sequencing Wins Over Static Playbooks

Evidence

- Hyper-Automation SOAR (paper 10): dynamic playbook generation beats static playbooks on novel incident types.
- GenDFIR (paper 20): two-stage rules + LLM pipeline outperforms either alone.

The agent must generate its own next-step hypothesis based on tool output, not follow a fixed script. Static sequencing fails on novel attack patterns -- exactly the category FIND EVIL targets.

3. Tool Output Grounding Is the Most Effective Hallucination Mitigation

Evidence

- Hallucination Survey (paper 24): 18 trigger causes identified; tool output grounding is most effective mitigation.
- ForensicLLM (paper 14): 86.6% source attribution -- every claim traceable to a tool output.
- GenDFIR (paper 20): pre-filtering via rules before LLM reasoning shrinks hallucination surface.

Every agent assertion must cite the specific tool call and raw output that produced it. Ungrounded narrative output is a disqualifying failure mode per the judging rubric (audit trail quality).

4. Use DFIR-Metric and Timeline Evaluation Framework for Benchmarking

Evidence

DFIR-Metric (paper 4): 700 questions, 150 CTF challenges, 500 NIST CFTT disk/memory cases -- first rigorous benchmark.
Timeline Methodology (paper 22): standardized evaluation framework for timeline reconstruction accuracy.

The accuracy report submission requirement is a judging criterion. Run DFIR-Metric questions during development. Score against known-good NIST CFTT cases. Use the timeline framework to quantify reconstruction accuracy before submission.

5. Agents of Chaos Failure Taxonomy + AIR Framework Define Negative Requirements

Evidence

Agents of Chaos (paper 16, [03_Resources/Articles/2026-03-08_Agents-of-Chaos](#)): 11 documented failure modes from live red-team autonomous agents.
AIR (paper 25): formal IR framework for agent systems -- rollback, containment, safe degradation.

The 11 failure cases are negative requirements: the submission must demonstrate it does not exhibit them. AIR framework defines how to handle the cases where the agent itself behaves incorrectly. Constraint implementation is a judging criterion -- architectural guardrails over prompt-based guardrails.

Cross-References

- [FIND EVIL! Hackathon](#) -- project index
- [03_Resources/Articles/2026-03-08_Agents-of-Chaos](#) -- paper 16, already analyzed
- [Hunt Evil Poster](#) -- baseline normal process knowledge
- [SIFT Cheatsheet](#) -- tool inventory the agent wraps
- [Windows Forensics Poster](#) -- artifact categories for agent triage logic
- [Zimmerman EZ Tools Poster](#) -- MCP candidate tool set